

Quiz 2 — Practice Problems

References: Unit 2: <https://github.com/wbandabarragan/computational-fluid-dynamics/tree/main/unit-2>; Munson, B. R., Rothmayer, A. P., Okiishi, T. H., & Huebsch, W. W. (2013). *Fundamentals of fluid mechanics* (7th ed.). John Wiley & Sons.

How to work through it

- Work in SI throughout, as in the lectures. Where the textbook poses a problem in British Gravitational units, convert the data first and answer in SI.
- Every time you write the Bernoulli equation, double check the four restrictions first — steady, inviscid, incompressible, along a streamline.
- Attempt each problem yourself before looking anything up. Getting stuck, and then unsticking yourself, is the way to learn.
- Estimate the order of magnitude of the answer before you compute it, and check it afterwards. This is an important step in understanding concepts and parameter spaces.
- Submission is not needed. This sheet is practice for Quiz 2.

Part A — Conceptual Questions

Read the *Conceptual Questions* on page 141 and answer **3.1C**, **3.2C**, **3.3C** and **3.4C**; then those on pages 189–190 and answer **4.1C**, **4.2C** and **4.3C**. Choose the correct answer and write 1–2 sentences justifying the choice, and say what makes each of the other options wrong.

Part B — Practice Problems

Solve the following problems from the textbook:

1. **Problem 3.2** — Bernoulli along a streamline
2. **Problem 3.6** — the pressure gradient needed to accelerate water.
3. **Problem 3.14** — the radial pressure gradient.
4. **Problem 3.19** — static, stagnation and dynamic pressure on a wing.
5. **Problem 3.24** — a Pitot-static tube measuring helium in a pipe.
6. **Problem 3.31** — a free jet from a faucet, and the effect of elevation.
7. **Problem 3.52** — continuity and Bernoulli equations
8. **Problem 3.87** — a siphon: the flowrate, and the pressure at the crown.
9. **Problem 3.116** — a Venturi meter under ideal conditions.
10. **Problem 3.126** — incompressible Bernoulli equation?
11. **Problem 4.3** — an unsteady velocity field.
12. **Problem 4.9** — the equation of the streamlines, and a sketch of them.
13. **Problem 4.12** — streakline of the steam.
14. **Problem 4.17** — the acceleration in circular motion.

15. **Problem 4.27** — local and convective acceleration.
16. **Problem 4.28** — average deceleration at shock wave.
17. **Problem 4.47** — the acceleration field of a 2D nozzle.
18. **Problem 4.60** — the material derivative of temperature.
19. **Problem 4.65** — The Reynolds Transport Theorem.
20. **Problem 4.69** — the control volume and the flux through each surface.

If you want to solve more: 3.11 and 3.13, the pressure gradient normal to a curved streamline; 3.58, a siphon on the edge of cavitation; 3.121, an inclined sluice gate; 4.11, a streamline against a streakline; 4.32, the acceleration field; 4.35, a stagnation streamline.